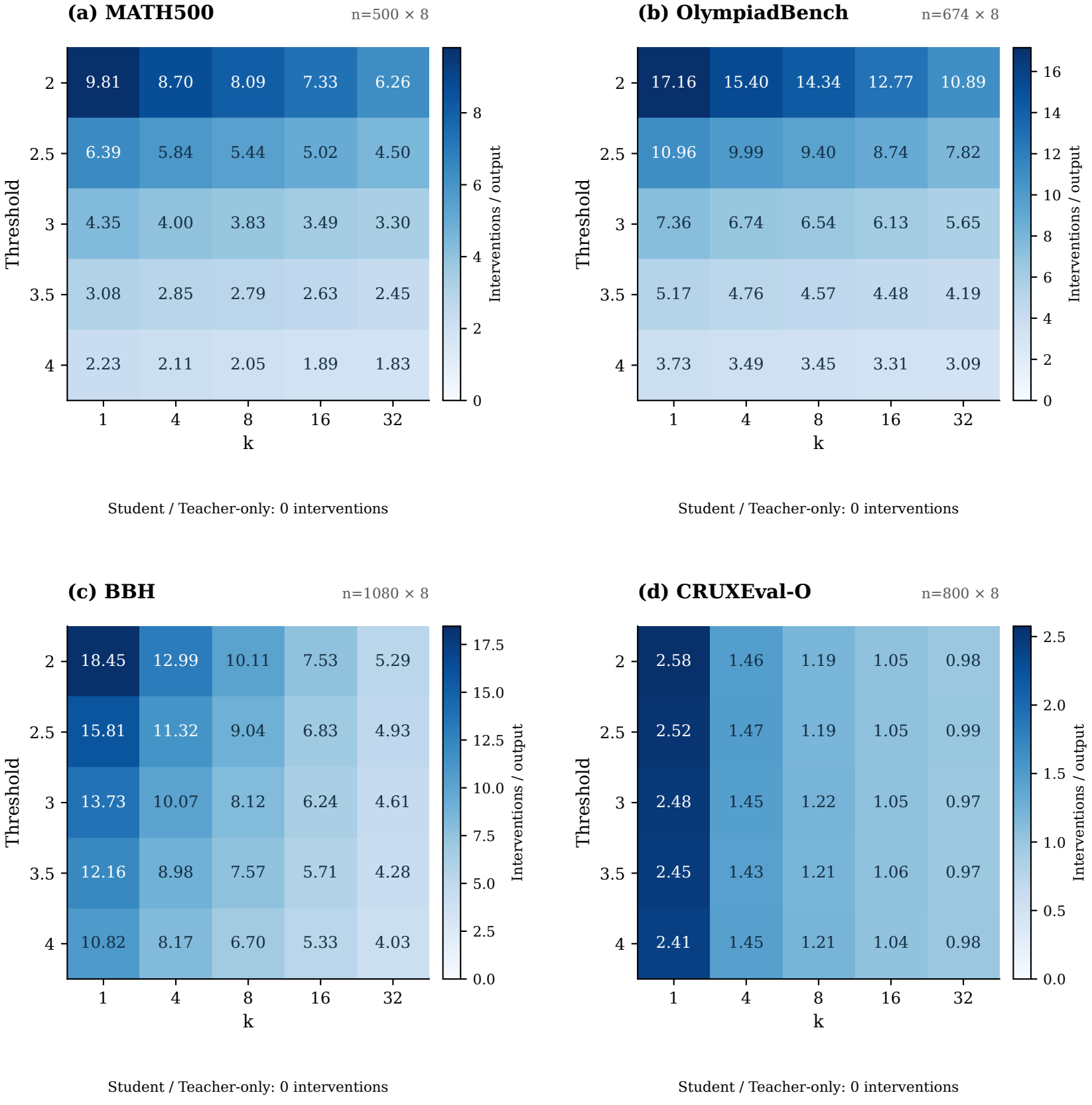

(a) MATH500
n=500 × 8
Threshold
k
Interventions / output
2 9.81 8.70 8.09 7.33 6.26
2.5 6.39 5.84 5.44 5.02 4.50
3 4.35 4.00 3.83 3.49 3.30
3.5 3.08 2.85 2.79 2.63 2.45
4 2.23 2.11 2.05 1.89 1.83
1 4 8 16 32
0 2 4 6 8
Student / Teacher-only: 0 interventions
(b) OlympiadBench
n=674 × 8
Threshold
k
Interventions / output
2 17.16 15.40 14.34 12.77 10.89
2.5 10.96 9.99 9.40 8.74 7.82
3 7.36 6.74 6.54 6.13 5.65
3.5 5.17 4.76 4.57 4.48 4.19
4 3.73 3.49 3.45 3.31 3.09
1 4 8 16 32
0 2 4 6 8 10 12 14 16
Student / Teacher-only: 0 interventions
(c) BBH
n=1080 × 8
Threshold
k
Interventions / output
2 18.45 12.99 10.11 7.53 5.29
2.5 15.81 11.32 9.04 6.83 4.93
3 13.73 10.07 8.12 6.24 4.61
3.5 12.16 8.98 7.57 5.71 4.28
4 10.82 8.17 6.70 5.33 4.03
1 4 8 16 32
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5
Student / Teacher-only: 0 interventions
(d) CRUXEval-O
n=800 × 8
Threshold
k
Interventions / output
2 2.58 1.46 1.19 1.05 0.98
2.5 2.52 1.47 1.19 1.05 0.99
3 2.48 1.45 1.22 1.05 0.97
3.5 2.45 1.43 1.21 1.06 0.97
4 2.41 1.45 1.21 1.04 0.98
1 4 8 16 32
0.0 0.5 1.0 1.5 2.0 2.5
Student / Teacher-only: 0 interventions